Network Security Laboratory

Experiment 6: Network Scanning Using Nmap

Stage	Fourth Year
Main Tools	Nmap / Zenmap + Python
Environment	Authorized isolated laboratory network only
Objective	Discover hosts, open ports, services, and simple security risks

1. Objective

Use Nmap to discover active devices, scan TCP ports, identify the service running on the target, and compare results before and after stopping a test service.

2. Required Setup

- Scanner Laptop: 192.168.50.20 /24
- Target Laptop: 192.168.50.10 /24
- Nmap or Zenmap on the Scanner Laptop
- Python on the Target Laptop
- Both devices on the same isolated laboratory network



Figure 1. Simple scanning topology

3. Start the Test Service

On the Target Laptop:

```
py -m http.server 8000
```

If py does not work:

```
python -m http.server 8000
```

This opens a test web service on TCP port 8000.

4. Perform the Scan



Figure 2. Simplified Nmap workflow

On the Scanner Laptop:

```
ipconfig
ping 192.168.50.10
nmap -sn 192.168.50.0/24
nmap -sV -p 1-10000 192.168.50.10
```

5. What the Commands Mean

Command / Option	Meaning
ping 192.168.50.10	Checks whether the target is reachable
-sn	Discovers active hosts without a port scan
-p 1-10000	Scans TCP ports from 1 to 10000
-sV	Attempts to identify the service and version

6. Expected Result

```
Nmap scan report for 192.168.50.10
Host is up
8000/tcp   open   http
```

The output may also identify the Python HTTP server version.

7. Interpret Port States

State	Meaning
open	A service is accepting connections
closed	The host is reachable, but no service is listening on the port
filtered	A firewall/filter prevents Nmap from determining the port state

An open port does not automatically mean compromise; it is an exposed service that should be required, updated, and protected.

8. Stop the Service and Rescan

On the Target Laptop:

```
Ctrl + C
```

Then on the Scanner Laptop:

```
nmap -sV -p 8000 192.168.50.10
```

Expected result:

```
8000/tcp   closed
```

This proves port 8000 was open because the test service was running.

9. Student Task

- Configure the scanner and target IP addresses.
- Start the Python web server on port 8000.
- Use -sn to discover active devices.
- Scan ports 1-10000 with service detection.
- Identify port 8000 and the detected service.
- Stop the Python service and rescan port 8000.

10. Required Screenshots

- Host discovery result.
- Open TCP port 8000 and detected service.
- Port 8000 after stopping the service.

11. Quick Troubleshooting

Problem	Check
Target not discovered	IP addresses, connection, ping, local firewall
Port 8000 not open	Confirm Python server is still running
Nmap command not found	Install Nmap or use Zenmap

12. Lab Questions

1. What is Nmap used for?

Discovering network devices, ports, and running services.

2. What does -sn do?

Discovers active hosts without scanning ports.

3. What does -sV do?

Attempts to identify the service and its version.

4. Which port is open in this experiment?

TCP port 8000.

5. How can open ports be secured?

Close unused services, update software, and configure the firewall.

Expected Outcome: Port 8000 appears open while the Python server is running and closed after the service is stopped.